KOHSUKE IDE

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Google Scholar
KohsukeIde

RESEARCH INTERESTS

Visual world models; self-supervised visual learning; 3D vision; vision-language-action models; diagnostic evaluation.

EDUCATION

4/2025 – 3/2027 (expected) **M.Eng. in Computer Science** — **University of Tsukuba, Japan**
Advisor: Prof. Yutaka Satoh. GPA: **4.27/4.30**.
Thesis (working title): *The Aftertaste of Progress: Rethinking Learning and Evaluation in Computer Vision.*

4/2021 – 3/2025 **B.Eng. in Applied Mechanics and Aerospace Engineering** — **Waseda University, Japan**
Advisor: Prof. Kiyoshi Saito. Minor in Computer Science.
Thesis (translated title): *Enhancing an Emulator-Based Load Testing System with Machine Learning.*

SELECTED PUBLICATIONS AND MANUSCRIPTS

1. **Kohsuke Ide**, Ryousuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, Yuki M. Asano, and Yutaka Satoh.
What Downstream Scores Miss in Point-Cloud Self-Supervised Learning.
Conference on Neural Information Processing Systems (**NeurIPS**), 2026 (Evaluations and Datasets Track).
Spotlight (top 1.28%; 48 oral/spotlight papers out of 3,757 submissions).
project

2. **Kohsuke Ide**, Ryousuke Yamada, Yue Qiu, Xianzheng Ma, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Beyond Single Object: Learning 3D Relations with Large Language Models.
IEEE/CVF Conference on Computer Vision and Pattern Recognition Findings (**CVPR Findings**), 2026.
project | paper | code

3. **Kohsuke Ide**, Ryousuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Seeing Red, Thinking Bad: Color Bias in Vision Language Models.
International Conference on Pattern Recognition (**ICPR**), 2026.
project | paper | code

4. **Kohsuke Ide**.
Consistency Is Not Identification: What Reuse Reveals About Latent Actions.
Under review at the International Conference on Learning Representations (**ICLR**), 2027.

5. **Kohsuke Ide**, Hiroki Nishizawa, Ryousuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Success Rates Are Not Model Properties: Controlled Diagnostics for VLA Evaluation.
Under review at the International Conference on Learning Representations (**ICLR**), 2027.

6. Hirokatsu Kataoka[*], **Kohsuke Ide**[*], Yoshihiro Fukuhara, Shumpei Takezaki, Nakamasa Inoue, Yuki M. Asano, and Christian Rupprecht.
What Have Vision Models Learned in the Deep Learning Era?
Under review at the International Conference on Learning Representations (**ICLR**), 2027.
[*] Equal contribution.

7. **Kohsuke Ide**, Ryousuke Yamada, Yue Qiu, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Structure First, Appearance Later: Label-Efficient Pretraining for Neural Radiance Fields.
Under review at the International Conference on 3D Vision (**3DV**) 2027.

8. Ryousuke Yamada, **Kohsuke Ide**, Yoshihiro Fukuhara, Hirokatsu Kataoka, Gilles Puy, Andrei Bursuc, and Yuki M. Asano.
3D sans 3D Scans: Scalable Pre-training from Video-Generated Point Clouds.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2026.
project | paper | code

9. Hirokatsu Kataoka, Yoshihiro Fukuhara, Yonglong Tian, Shangzhe Wu, Oishi Deb, Ryousuke Yamada, Christian Rupprecht, Jianyuan Wang, **Kohsuke Ide**, Koichi Namekata, Xianzheng Ma, Yiming Chen, Robert Geirhos, Aditi Raghunathan, Yuki M. Asano, Deva Ramanan, David Fouhey, Andrew J. Davison, Yilun Du, Jiajun Wu, and Zhuang Liu.
Visual General Intelligence: A White Paper.
Preprint, 2026.
paper

10. **Kohsuke Ide**, Ryousuke Yamada, Yoshihiro Fukuhara, Hirokatsu Kataoka, and Yutaka Satoh.
Colors You Can't See, Semantic Biases You Can't Ignore.
ICCV Workshop on Multimodal Reasoning and General Intelligence (**ICCV Workshop**), 2025.

RESEARCH EXPERIENCE

9/2025 – Present ***Research Assistant*, AIST** **Tsukuba, Japan**

- Conduct research on visual world models, 3D understanding, video representation learning, and vision-language-action models at the National Institute of Advanced Industrial Science and Technology (AIST).
- Lead projects from problem formulation through implementation, experiments, and writing.
- Collaborate with Prof. Yuki M. Asano on video representation learning, and with researchers at Oxford's Visual Geometry Group on 3D and video understanding.

1/2024 – 8/2025 ***Research Intern*, AIST** **Tsukuba, Japan**

- Developed diagnostic evaluations for shortcut learning, language leakage, and visual biases in 3D and vision-language models.
- Built multi-object 3D reasoning benchmarks that require differences between scenes rather than language-only cues.
- Designed and implemented a video-to-point-cloud data-generation pipeline for scalable 3D pre-training.

8/2024 – 3/2025 ***Research Intern*, Preferred Networks** **Tokyo, Japan**

- Developed and improved 3D reconstruction and free-viewpoint video pipelines.
- Optimized rendering and model components for efficient experimentation.

9/2021 – 3/2024 ***Machine Learning Engineer Intern*, Lightblue Technology** **Tokyo, Japan**

- Developed and deployed an object-tracking system for a work-efficiency monitoring application using Docker and AWS EC2.
- Improved human-action-recognition accuracy by 20% through feature engineering and Optuna-based hyperparameter search.

9/2023 – 10/2023 ***MLOps Engineer Intern*, M3** **Tokyo, Japan**

- Built **RecoverOOM**, an open-source service that detects out-of-memory failures in GKE CronJobs, updates memory limits, and restarts jobs automatically.

RESEARCH VISITS & INTERNATIONAL COLLABORATION

19–31 Jan 2026 **Research Visit** **University of Oxford, UK**
Research meetings with members of the Visual Geometry Group and Active Vision Laboratory.

22–30 Apr 2026 **ASPIRE Computer Vision Workshop and Research Visit** **University of Oxford, UK**
Presented research and discussed follow-up work with researchers at Oxford VGG.

INVITED TALKS

- **Seeing What Matters, Learning How to Find Out: Toward Visual General Intelligence.**
  CMU Video Model Journal Club (online), 18 Sep 2026 (PT). recording

ACADEMIC SERVICE

- **Reviewer**, International Conference on Learning Representations (ICLR), 2027.
- **Workshop Organizer**, **Visual General Intelligence: Vision Research Toward the AGI Era**, CVPR 2026.
- **Organizer**, **LENS (Young Researchers in Vision)**
  A Japanese academic community of over 400 members ranging from high school students to professors.

HONORS & AWARDS

- **MIRU 2025 Interactive Presentation Award** (Jul 2025)
  Selected as one of **11** award recipients from **601** interactive presentations (**top 1.8%**) for "Can 3D Large Language Models Count to Three?"
- **WINC Hackathon Winner** (2023)
  Built a ResNet-based push-up counter optimized with TensorRT for Jetson Nano.

TECHNICAL SKILLS & LANGUAGES

- **Programming and systems:** Python, C++, Go, PyTorch, CUDA, TensorRT, OpenCV, Git, Docker, Kubernetes, Linux, AWS, and GCP.
- **Languages:** Japanese (native); English (fluent; TOEFL iBT 118/120, TOEIC 990/990).